

# Helix OTA “ Continuation / Resume Handoff

| Field          | Value                                                                                                                                                                                                                                                                                |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Revision       | 3                                                                                                                                                                                                                                                                                    |
| Created        | 2026-06-07                                                                                                                                                                                                                                                                           |
| Last modified  | 2026-06-10                                                                                                                                                                                                                                                                           |
| Status         | active “ resume with “continue”<br>Single source of truth for resuming work.<br>Captures exactly what is DONE (verified), the git state, and the prioritized NEXT steps.<br>Everything below is committed to main and pushed to all 4 upstreams (GitHub, GitLab, GitFlic, GitVerse). |
| Status summary |                                                                                                                                                                                                                                                                                      |

## How to resume

Type `continue` in a new session. Read this file + the memory index first. All work is on `main` (latest commit pushed to all upstreams). Working branch history: `foundation` “ `research` “ `MVP specs` “ `submodule impls` “ `server`, each merged to `main`.

## Locked decisions (do not re-litigate)

- Stack: **Go + Gin + Brotli + HTTP/3(QUIC)** “ **HTTP/2 fallback, REST primary**; reuse `basic-digital/http3`.
- Strategy: **native Android A/B** (`update_engine` + **AVB/dm-verity** + **auto-rollback**) + **custom Go control plane**; wrap OSS only where it adds value (hawkBit gated, ADR-0001).
- Trust: signing + SHA-256 + AVB for MVP; TUF server-side then device-side in 1.0.1 (ADR-0002).
- Topology: modular monolith for MVP (ADR-0003).
- New submodule repos auto-created PUBLIC on GitHub + GitLab (pre-authorized).
- `docs/research/main_specs/additions/` files are authoritative input “ always analyze + fold in.
- Commit + push to ALL upstreams regularly. Merge to `main` when a milestone is done.

## Session 2026-06-08 update (DONE, on main, pushed all 4 upstreams)

- **Constitution submodule relocated** `HelixConstitution/` “ `constitution/` (repo/URL unchanged) via `git mv`; fixed the only filesystem path refs (`tests/test_strategy.md`, `additions/initial_research_01.md`) incl. “ their pre-existing off-by-one link depth. Prose “`HelixConstitution`” clause citations (the repo NAME) intentionally left.
- **Inheritance wired**: parent `CLAUDE.md`, `AGENTS.md`, `docs/guides/HELIX_OTA_CONSTITUTION.md` inherit from the submodule; `tests/`

- inheritance\_gate.sh (5 invariants, matches the exact Â§11.4 forensic-anchor heading) + tests/test\_constitution\_inheritance.sh (gate green clean + Â§1.1 paired-mutation proven via constitution/meta\_test\_inheritance.sh + recursive submodule pointer check) + tests/pre\_build\_verification.sh + installed pre-commit hook. All 6 owned ota-\* submodules carry CLAUDE.md+AGENTS.md inheritance pointers (pushed github+gitlab; gitlinks bumped to v0.1.0-1-gâ€¦). Commits ee5dc7d, e11a221.
- **All 3 additions processed** (operator mandate â€” nothing skipped): per-input exhaustive inventories in research/additions\_analysis/0{1,2,3}\_analysis.md; research/additions\_synthesis.md â†’ **Rev 2** with consolidated 14-gap register (Â§8), new-work routing (Â§9), addition-#3 conflict reconciliation (Â§10, locks Gin/ed25519/JWT/releases+deployments/â‰¥90%), future-phase catalogue (Â§11), UNVERIFIED register (Â§12). MVP-critical **G1 (anti-downgrade invariant)** implemented as regression tests; handleClientUpdate 79.3%â†’86.2%. Commit 629b4eb.
  - **Constitution path is now constitution/** â€” update any new references accordingly.

## Server hardening â€” DONE this session (handoff NEXT #1, real evidence)

- **Coverage:** upload handler 71.2%â†’**96.6%**, client-update 79.3%â†’**93.1%** (â‰¥90% safety floor). Commit 64c3bfd.
- **pgx PostgreSQL Repository** (server/internal/store/postgres.go): full store.Repository impl; shared contract\_test.go proves parity with memory; **integration test boots real Postgres via the containers submodule on podman** (go test -tags integration ./internal/store/), evidence in docs/qa/20260608-pgx-postgres-integration/. Surfaced+fixed a real idempotency overwrite bug. Commit 96fdecb.
- **Brotli+gzip+identity** compression middleware (server/internal/api/compression.go). Commit b26f30e.
- **HTTP/3 (QUIC)+HTTP/2 fallback** via the new submodules/http3 (digital.vasic.http3) â€” server/internal/transport/; wired into cmd/ota-server (TLS via HELIX\_TLS\_CERT/KEY â†’ HTTPS port 8443; plain HTTP otherwise). Real h3+h2 client test; evidence in docs/qa/20260608-http3-h2-brotli-transport/. Commit 1469edf.

## Parallel-wave deliverables â€” DONE this session (commit 8efb6b8)

6 subagents closed additions gaps: **G2** validator hash-before-signature VERIFIED (FACT, file:line); **G7** staged-rollout engine spec + migration\_002 design; **G6** dashboard design; **G3/G4/G5** operational endpoints spec (audit/telemetry-reads/group CRUD + proposed repo methods); **G10** CI (.github/workflows/ci.yml + CODEOWNERS + dependabot); **Â§11** future-phase folding (1.0.2-rollback, 1.0.3-delta-updates created; 1.X-linux/windows/other-os extended).

## Operational endpoints â€” DONE this session (G3/G4/G5, real-Postgres-verified, memory+pgx parity)

- **G3 audit** (commit 9a703bf): auditMiddleware records successful mutating actions (reads/failures skipped, ids never leak into the action) + GET /audit (admin) + audit\_logs table.
- **G4 telemetry reads** (commit eadaa7f): GET /devices/{id}/telemetry (device reads only its own) + GET /telemetry/overview (fleet counts).
- **G5 device-group CRUD** (commit e3e1307): full /groups CRUD + membership; writes operator/admin, delete admin-only. All three added store.Repository methods on

memory + pgx, extended the shared contract, and pass the pgx integration test on real Postgres.

## Staged rollout + more " DONE this session

- Migration 002 real-DB validated (e14942a). **Go engine** wired via the ota-rollout-engine brick: server/internal/rollout + REST POST/GET / deployments/{id}/rollout + /evaluate (create"start"advance"complete, halt-on-error-breach); 7 tests green (046ea07).
- End-user rollback reconciled INTO **1.0.1** per operator decision; 1.0.2-rollback superseded"folded (a4c2b3d).
- **G11** repo audit: all 6 ota-\* PUBLIC on GitHub+GitLab (dab2f0e). **G12** NFR/load harness real measured percentiles (6547c7f).

## Round 2 deliverables " DONE this session

- **Dashboard** scaffold (G6) " Vite+React+TS, real tsc/vite build exit 0 (813a00e).
- **6 submodule READMEs** enriched + pushed to each github+gitlab (813a00e).
- **Dependabot "4 merged + verified:** gin 1.12.0, quic-go 0.60.0 (http3 transport re-verified), actions v6 (33039b8).
- **Rollout pgx StoragePort** " real-DB tested via containers submodule (8190e92).
- **"11.4.65 exports** for 13+2 new docs (242ee2f, a288a4b). **device\_tuf.md** + **rollback\_ux.md** specs (1.0.1).

## Round 3 deliverables " DONE this session

- **Rollback-history store layer** (AppendRollback/ListRollbacks) memory+pgx, real-DB parity (e4165a6).
- **Recall endpoint** POST /deployments/{id}/recall + GET /rollbacks (c98dfac) " records rollback\_history, validates deployment/target-release, operator/admin; 3 tests.
- **Delta-updates 1.0.3** design (ADR-0005 Option B) + **as-built endpoint reference** (server.go route table) + submodule README "11.4.65 siblings (573779a).
- **Autonomous e2e challenge tests/e2e/challenge\_operational.sh** " real live-server run, 28 passed/0 failed/1 skip, independently re-verified (bb332a4).

## Round 4 deliverables " DONE this session

- **pgx wired into main** (HELIX\_DATABASE\_URL): production uses the pgx repo + rollout StoragePort; **e2e challenge PASS (28/0/1) against a real Postgres-backed server** (1cdad81, docs/qa/20260608-pgx-server-e2e/).
- **migration 004** delta\_artifacts real-DB validated; **CI** e2e+loadtest jobs; **README** doc-map "10; **HelixQA** bank; **threat-model** extended (6a3e213, c-threat).

## Round 5 deliverables " DONE this session

- **Recall = forward-fix** (operator decision honor-AVB, 4e35c3e): handleRecall supersedes the current deployment + creates a NEW active deployment of the target release; the update-check anti-downgrade invariant means AVB is honored by construction. New store UpdateDeployment (memory+pgx, real-DB parity). Decision recorded in rollback\_ux.md Rev 2 + threat\_model.md "11.11 RESOLVED (f5ec504).

## Round 6 deliverables – DONE this session

- **Delta-artifact store + API** (66464d7): `delta_artifacts` (base% target CHECK + UNIQUE pair) on memory+pgx; POST/GET /deltas register+lookup; real-DB parity.
- **device-TUF client-decision memo** (recommend gomobile-go-tuf/v2, ADR-0002 §4.3) + sibling.
- **Additive WIDENs – full set landed** (operator-approved): `audit ?since/?until` filters + `telemetry failure_rate/by_state` (028e656, memory+pgx real-DB parity) + `group member_count` (4cb86d7). `spec_impl_alignment.md` Rev 3: rows 2+5+6-partial landed; as-built doc re-synced (recall now WIRED, /deltas + recall + rollbacks documented).

## Round 7 – breaking WIDENs landed (operator ruling: WIDEN-impl)

- **Group** (a91271b + 02ad2d0): `id group_id`; batch member-add 200 {added, already\_member, not\_found}. Server+tests+e2e(36/0/1)+dashboard+bank+docs.
- **Audit** (fbaefbe): actor object {user\_id, subject}. Server+test+dashboard+docs.
- **Telemetry per-device** (2a48ab5): `events items`, newest-first, ?limit/?cursor+next\_cursor. Server+tests+dashboard+docs.
- **GET /members** (3b4b1d8 + e2e): `device_ids[] items[]` of {device\_id, added\_at}; store gained `device_group_members.added_at` + `ListGroupMembersDetailed` (memory+pgx, real-DB parity); e2e 39/0/1.
- Each: full default suite green, all wire consumers updated in lockstep, `spec_impl_alignment.md` rows 6/8/1/4-structural + GET-members RESOLVED.

**WIDEN ruling status: COMPLETE** except two legitimately-parked items: row-4 richer telemetry fields (`duration_ms/bytes_transferred`) **blocked on UNVERIFIED ingest** (event source must carry them first) + the per-device telemetry filters; group/members list pagination (row 7) **deferred** (groups bounded – memo™s own recommendation). Both need either ingest work or an operator nudge; not autonomously actionable now. *(UPDATE 2026-06-10: the per-device telemetry filters + group/members list pagination are now DONE – shipped 50ef5c6; see the –Session 2026-06-10 update– below. Only row-4 richer numeric fields remain parked on ingest.)*

## Round 8 – all of it – wave (operator: do all frontiers)

- **Server-side delta-selection** (9179d0a): `ota-protocol UpdateAvailable.Delta` (brick 7d18edc, pushed; server builds via dev replace); store `ReleaseByVersion` (memory+pgx, real-DB parity); update-check resolves current base artifact FindDelta offers delta + full fallback; `TestClientUpdateOffersDelta`.
- **Device-side delta-apply** (ota-android-agent, pushed + gitlink): pure `DeltaApplyDecision` (USE\_DELTA vs FULL\_PAYLOAD), 11 tests, :core 47/0 real Gradle.
- **Operator cleanup** (DONE): vasic-digital/containers + HelixConstitution GitLab mirrors flipped private→public (re-read-verified); CODEOWNERS @milos85vasic confirmed.
- **OpenAPI** synced to all widened/new endpoints + `UpdateAvailable.delta`; §11.4.65 corpus export coverage (46 html+pdf pairs).

## Round 9 “ hardware-free QA wave (operator: “all of it”, no Orange Pi tonight)

- **Stress + chaos** (Â§11.4.85, 5363511): race-clean in-process suite “ 200 concurrent creates (0 err, p99 6.4ms), 2400 sustained reads (0 err), 60-device membership contention “ no lost updates, chaos fault “ 500 “ recover “ 200. Evidence docs/qa/20260608-stress-chaos/.
- **Full-pipeline e2e + security** (e7e1alc): tests/e2e/pipeline\_signed.sh (real ed25519-signed artifact “ upload “ release “ deploy “ rollout “ delta-bearing update; bogus-sig “ 422) 32/0/0 “ **closes the artifact-pipeline SKIP**; tests/security/security\_probes.sh (authn/authz/ownership/injection/trust-boundary) 37/0/0. Both re-run in-tree.
- **Performance/NFR + scaling** (7b75212): real memory-vs-pgx concurrency sweep “ in-mem 20.5k RPS @ c=128 p99 30.6ms; pgx ~6.2k plateau (2-CPU container). docs/research/main\_specs/1.0.0-mvp/nfr/performance\_baseline.md.
- **Dashboard build-out + Playwright** (just landed): functional Fleet/Deployments+Recall/Groups/Audit screens on the real API; tsc+build exit 0; **Playwright 5/5** vs a live server.
- **Coverage ledger** (Â§11.4.25) + **HelixQA bank** now 8 challenges (incl.Â signed-pipeline + security).
- **Benchmarking** (Â§11.4.27, bbc97e4): Go testing.B suite “ healthz 2.5Âµs, group-create 8.5Âµs, update-check 4.9Âµs, FindDelta 160ns/0-alloc, etc. docs/qa/20260608-benchmarks/.
- **DDoS/flood probe** (Â§11.4.27, 6d29fb8): 5,952-req burst served, responsive post-flood; surfaced the honest “no rate-limiter“ finding.
- **Rate-limit FEATURE** (1b97fc7): implemented the finding “ “ fix “ in-flight cap middleware (HELIX\_MAX\_INFLIGHT, default-off) sheds 429 RATE\_LIMITED; proven cap=1/300-concurrent “ 244 shed / 56 served / recovers. **Â§11.4.27 test-type matrix now complete** for the operational surface (unit/integration/e2e/security/ddos/scaling/chaos/stress/performance/benchmarking/ui/Challenges). Full rebuild+validate sweep green (docs/qa/20260608-full-rebuild/).

## Session 2026-06-10 update (DONE, on main, pushed all 4 upstreams)

- **Per-device telemetry filters + group/members pagination** (50ef5c6, feat(api) : per-device telemetry filters + group/members pagination): closes the two previously-PARKED WIDEN bits “ per-device telemetry now accepts the filter params, and the group/members list is paginated. **OpenAPI synced + redocly-clean**. These move out of the “parked WIDEN bits“ list (no longer deferred).
- **Dashboard client lockstep** (b0b8ee2, chore(dashboard) : sync API client to new pagination/filter params): the dashboard API client updated in lockstep with the new pagination/filter params, so every wire consumer stays consistent (Â§11.4.92 cross-feature consistency). **HEAD is now b0b8ee2**.
- **Emulator-driven device testing initiative started** “ tiered plan captured as FACT in docs/design/EMULATED\_DEVICE\_TESTING.md. **Tier-1 (podman ota-device-emulator over real ota-protocol) is IN PROGRESS**; Tier-2 (Cuttlefish A/B, Linux+nested-KVM-gated) and Tier-3 (real RK3588, hardware-gated) are designed, host/hardware-gated (NOT structurally impossible per Â§11.4.112). Extends the containers submodule (pkg/boot/compose/health + pkg/emulator AVD-x86\_64 + pkg/vm qemu-aarch64) per Â§11.4.76.
- **Canonical Â§11.4.131 session-resumption file created** at docs/RESUMPTION.md “ the fixed out-of-the-box entry point for any fresh session (SHORT + FULL variants, read-

first handoff pointers, live-state anchors, PHASE/NEXT/terminal-goal, binding constraints). Point a new session at that one file.

## NEXT wave (still open â€” all hardware/ingest-gated)

1. **Device-side TUF** (gomobile-go-tuf/v2 per the decision memo) â€” gated on an arm64 .so-size/JNI measurement on real RK3588 hardware.
2. **Device payload-apply integration** â€” wire `DeltaApplyDecision` into the on-device apply path (`:android/update_engine`) â€” needs a real device to validate end-to-end.
3. **Emulator-driven device testing** â€” tiered plan now in flight (`docs/design/EMULATED_DEVICE_TESTING.md`): **Tier-1 IN PROGRESS** (podman ota-device-emulator speaking real ota-protocol to the control plane â€” registerâ†’update-checkâ†’telemetryâ†’deltaâ†’rolloutâ†’recall, runnable on this macOS host now); Tier-2 Cuttlefish A/B (Linux+nested-KVM-gated); Tier-3 real RK3588 (hardware-gated).
4. **Parked WIDEN bits**: row-4 richer telemetry fields (`duration_ms/bytes_transferred`) â€” still blocked on UNVERIFIED ingest (event source must carry them first). **Telemetry per-device filters + group/members list pagination are now DONE** (shipped this session, see below) â€” no longer parked.

## Carried-forward gaps register

See `additions_synthesis.md` Â§8/Â§9 (14 gaps; most now specced â€” implementation pending). Numbering decision: 1.0.1 = staged-rollout; rollbackâ†’1.0.2, deltaâ†’1.0.3.

## DONE (verified, on main, pushed)

1. **Spec corpus** (`docs/research/main_specs/`): master design + ADRs index; research = scored landscape report + 12 stack notes + 5 ADRs (ADR-0001..0005) + additions synthesis; foundation = glossary, STRIDE threat model, submodule reuse map, doc standards, requirements traceability, export pipeline; **1.0.0-mvp** full specs (api+OpenAPI, database+migrations, security, server, client\_android+snippets, deployment, tests, `VALIDATION_EVIDENCE`); 6 Mermaid diagrams; future-phase outlines (1.0.1, 1.X-linux/windows/other-os); corpus README.
2. **Validation evidence (real)**: OpenAPI redocly-valid; **migrations applied to live PostgreSQL** (12 tables up, clean down); k8s kubeconform 5/5; compose YAML valid; all 6 diagrams render (mmdc); corpus exports = 50 HTML + 50 DOCX + SVG/PNG (`scripts/export_docs.sh`). PDF needs a LaTeX engine (not installed).
3. **6 reusable submodules** (created on GitHub `HelixDevelopment/` + GitLab `helixdevelopment1/`, scaffolded, wired under `submodules/`, **implemented v0.1.0, tagged, pushed both hosts**):
  - ota-protocol (Go, 99.3%), ota-telemetry-schema (Go, 98.9%), ota-artifact-validator (Go, 97.8%, ed25519+sha256), ota-rollout-engine (Go, 94.9%, halt-wins+deterministic cohorts)
  - ota-update-engine-bridge (Kotlin; :core 27 tests; :android builds an AAR)
  - ota-android-agent (Kotlin; :core 36 tests; :android build pending KMP/AGP-on-Gradle-9.5 alignment)
4. **Submodules wired**: HelixConstitution + containers at root; 6 ota-\* under `submodules/`.
5. **Control-plane server** (`server/`, Go module `github.com/HelixDevelopment/helix_ota/server`): Gin modular monolith wiring the 4 Go modules; full /api/v1 MVP endpoints (auth, devices, artifacts upload+validate S1â€”S6, releases, deployments all-targets, client/update 204|200, telemetry, health); **66 httptest integration tests pass**; gofmt/vet/build clean.

## NEXT (prioritized)

1. **Server hardening to spec:** raise internal/api coverage to ~90% on `handleUploadArtifact + handleClientUpdate` (error-path branches); add a **pgx** Repository implementation behind the existing `store.Repository` interface + `testcontainers-go` integration tests against real Postgres; wire `vasic-digital/http3` for HTTP/3 + HTTP/2 + Brotli.
2. **ota-android-agent** : **android build:** switch its `:android` module from Kotlin MPP to `plain com.android.library + org.jetbrains.kotlin.android` (like the bridge, which builds), OR pin Gradle 8.x + AGP 8.7.x via a wrapper; then produce the AAR.
3. **React dashboard** (`dashboard/`): reuse `UI-Components-React`, `Dashboard-Analytics-React`, `Auth-Context-React`; secure login, upload, deploy, fleet health.
4. **1.0.1 staged-rollout:** promote the outline to full specs + implement rollout API (phases, pause/resume/abort), device-side TUF, end-user rollback; migration `002_*` (`deployment_phases`, `rollouts`, `rollback_history`, TUF metadata).
5. **AOSP integration:** build the agent + bridge as a system/priv-app in the Orange Pi 5 Max AOSP tree; on-device e2e (`download` + `verify` + A/B `apply` + `reboot` + `post-boot`; `corrupt-slot` + `auto-rollback`).
6. **Containerize** the dev stack via the `containers` submodule; CI running all validators (`redocly`, Postgres migrations, `kubeconform`, `go test`, `gradle :core:test`, `export render`).
7. **Per-submodule:** full README/docs/manuals + GitFlic/GitVerse mirrors + tag mirroring (§4).

## Toolchain facts (this host)

Go 1.26.2; Android SDK (platforms 31~36.1, build-tools, NDK); Gradle 9.5 + Kotlin 2.3.20; **AGP 8.5.2 + plain kotlin.android works on Gradle 9.5; Kotlin MPP does NOT** (needs Gradle 8.x). `pandoc/mmdc` present; no `LaTeX/drawio/plantuml/graphviz/docker`. Live PostgreSQL server available. `gh+glab` authed. API rate limits are transient ~ keep waves ~6 agents, don't overlap workflows.